

HerHealthEval: Do Language Models Understand Women’s Health Communication Across Languages and Cultures?

Anonymous ACL submission

Abstract

Large language models are increasingly used to answer women’s-health questions, but fluent answers do not establish that a model has correctly understood what a patient is trying to communicate. We present HerHealthEval, a multilingual benchmark and evaluation framework for studying language understanding in women’s-health communication across languages and linguistic registers.

Our controlled evaluation suite covers menstruation, polycystic ovary syndrome (PCOS)/hormonal concerns, and fertility in English, Arabic, and French, pairing clinical, lay, indirect, ambiguous, and emotionally concerned expressions with predefined targets for concern category, risk level, recommended action, and clarification need. The protocol requires native-speaker validation of Arabic and French for meaning preservation, naturalness, and register before cross-lingual evaluation or adaptation, while evaluation cases remain separated from leakage-controlled adaptation data. The experimental design centers Qwen3.5-9B and specifies a general-purpose baseline, English domain adaptation, and subsequent Arabic/French adaptation, evaluated through misunderstanding, under-triage, clarification, unsafe-response, and cross-style/cross-language consistency measures. Empirically, we find that naive multilingual fine-tuning can *degrade safety below the base model*: a fine-tune that interprets French and Arabic as well as the base model nonetheless under-triages almost every non-English case (under-triage ≈ 1.0 vs. the base model’s 0.42–0.46), a failure a controlled ablation traces to language-dependent label quality rather than comprehension (paired risk discordance entirely one-directional, $p < 10^{-63}$). Correcting the labels to be language-independent restores non-English safety to a level statistically indistinguishable from the base model ($p = 0.88/0.94$ for French/Arabic) while yielding the best interpretation accuracy of

any configuration. Rather than claiming comprehensive clinical coverage, this preliminary study establishes a failure-oriented protocol for testing women’s-health language understanding beyond answer generation.

1 Introduction

The central question in this work is not whether a large language model (LLM) can produce a plausible answer to a women’s-health question, but whether it first understands what the patient is trying to communicate. The same concern may be stated clinically (“I have irregular menstrual bleeding”), colloquially (“my period is all over the place”), or indirectly (“my monthly problem has changed”). Although these expressions can point to the same underlying concern, they differ in explicitness, emotion, and cultural register. Misreading that variation can lead a model to confuse the concern, understate urgency, offer false reassurance, or answer when it should first ask for missing information (Recker et al., 2025; de Vries et al., 2025).

Most women’s-health LLM studies, however, evaluate the generated answer: its accuracy, readability, empathy, or user preference. For example, MenstLLaMA evaluates domain adaptation for menstrual-health education, while recent expert-led women’s-health evaluations probe response safety across clinical scenarios (Adhikary et al., 2025; Maurya et al., 2026). These studies are important, but they do not isolate whether a matched clinical intent survives changes in surface form. Multilingual health QA adds a second concern: models can give inconsistent health information when the same question is posed in different languages (Schlicht et al., 2025). Work from LUHME similarly shows that LLMs may compress cross-cultural variation, that sociocultural context affects how meaning should be adapted,

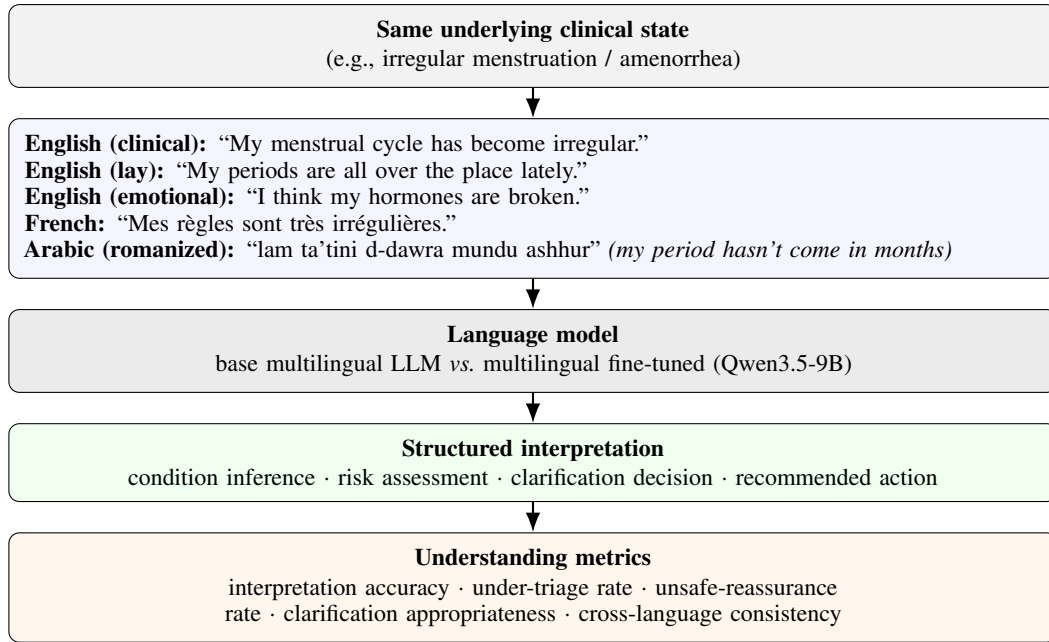


Figure 1: **HerHealthEval overview.** The same underlying clinical state is expressed across languages (English, French, Arabic) and communication registers (clinical, lay, emotional, indirect/cultural, ambiguous). The central question is whether a model *understands what the patient means*—inferring condition, risk, and the need for clarification—*before* deciding how to respond. We measure this with language-understanding metrics (interpretation accuracy, under-triage / unsafe-reassurance, clarification appropriateness, and cross-language consistency) rather than answer accuracy.

and that successful dialogue depends on establishing common ground (Mohammadi et al., 2025; Freitas and Cardoso, 2025; Anikina et al., 2025). Together, these findings motivate a controlled evaluation of *interpretation* rather than another medical answer-generation study.

We therefore present **HerHealthEval** (Figure 1), an initial investigation of language understanding in multilingual women’s-health communication. The controlled evaluation suite spans menstruation, polycystic ovary syndrome (PCOS)/hormonal concerns, and fertility; it holds the underlying case fixed while varying canonical, clinical, layperson, indirect/culturally sensitive, ambiguous, and emotionally concerned expression. Matched English cases are localized into Arabic and French under a protocol that requires native-speaker approval of semantic preservation, naturalness, and register before they enter cross-lingual evaluation or adaptation. Each item has predefined targets for concern category, risk level, recommended action, and clarification need. The evaluation lineage is frozen separately from the adaptation corpus, with exclusions based on both source provenance and paraphrase-family identity.

We organize the study around three research

questions:

1. **RQ1:** How robust are LLM-based clinical reasoning systems to variation in linguistic register when interpreting semantically equivalent women’s health concerns?
2. **RQ2:** Do multilingual LLMs exhibit systematic differences in interpretation, triage calibration, and clarification behavior across English, Arabic, and French?
3. **RQ3:** What is the impact of domain-specific and multilingual adaptation on linguistic robustness, uncertainty handling, and clarification generation in women’s health symptom understanding?

The contributions align with these questions: (1) a formulation of women’s-health interaction as a controlled language-understanding problem across register; (2) a multilingual evaluation framework with matched language variants, a native-speaker validation gate, and leakage-separated adaptation data; and (3) a reproducible, failure-oriented protocol for comparing general-purpose and sequentially adapted models using misunderstanding, triage, clarification, safety, and

consistency measures. These contributions are deliberately preliminary: the study establishes an evaluation protocol and supports analysis of early failure patterns rather than claiming comprehensive coverage or clinical readiness.

The remainder of this paper reviews related work (§2), describes the evaluation suite and experimental protocol (§3), presents results and error analysis (§4), and discusses limitations and ethical considerations (§5).

2 Related Work

We cluster the closest work into three lines: women’s-health answer evaluation, multilingual and sociocultural language understanding, and evaluation design for ambiguity and safety.

2.1 Women’s-Health Answer Evaluation

Women’s-health LLM research primarily asks whether generated answers are accurate, readable, empathetic, or safe (Recker et al., 2025; de Vries et al., 2025). Adhikary et al. (2025) introduce MenstLLaMA, the closest domain-adaptation system, by fine-tuning LLaMA-3-8B on menstrual-health question–answer data and evaluating educational response quality in English. Mughal et al. (2025) develop Mai, domain-specific menstrual-health dialogue systems in English and Roman Urdu; they explicitly motivate adaptation through local language, stigma, religious norms, and sociocultural barriers, but evaluate generated dialogue through automatic metrics and user ratings rather than matched tests of interpretation. For PCOS, Devranoglu et al. (2024) evaluate ChatGPT on clinician-authored questions about PCOS-related infertility, and Graca et al. (2026) conduct an international blinded survey of health-professional ratings of ChatGPT and evidence-based self-management responses. For fertility, Chervenak et al. (2023) compare ChatGPT counseling answers with public-health and professional sources, while Gély et al. (2026) compare LLM answers to infertility FAQs with specialist answers on quality, accuracy, and empathy. Maurya et al. (2026) broaden women’s-health coverage through WHBench, using expert-crafted scenarios and a safety-aware rubric. Together, these studies motivate domain-specific adaptation and careful response evaluation, but they generally treat the input question as fixed; they do not test whether the same concern is interpreted consistently when its

register or language changes.

2.2 Multilingual and Sociocultural Understanding

Jin et al. (2024) introduce XLingEval to assess correctness, consistency, and verifiability on naturalistic healthcare questions across English, Spanish, Chinese, and Hindi, finding substantial cross-language disparities. Subsequent multilingual health QA work likewise reports that equivalent questions can elicit inconsistent information across languages (Schlicht et al., 2025). Foundational cross-cultural NLP work distinguishes linguistic diversity from cultural variation and identifies linguistic form, common ground, topic relevance, and communicative objectives as separate dimensions of culturally situated language use (Hershcovich et al., 2022; Adilazuarda et al., 2024). In reproductive health specifically, participatory work with underserved women shows that culturally appropriate responses require local context beyond literal translation (Deva et al., 2025), while evaluation rubrics built around Western assumptions can underrate responses judged appropriate by trained annotators and public-health experts (Dey et al., 2025). The LUHME literature identifies related problems: models compress observed cultural variation (Mohammadi et al., 2025), text adaptation can fail when sociocultural context is ignored (Freitas and Cardoso, 2025), and dialogue systems require mechanisms for establishing common ground (Anikina et al., 2025). Cross-register transfer work further motivates holding semantic content constant when measuring sensitivity to linguistic form (Nicholls and Alperin, 2025). Our setting brings these concerns together in matched English, Arabic, and French women’s-health descriptions rather than testing general medical knowledge alone.

2.3 Gap and Evaluation Requirements

The resulting gap is an evaluation framework that separates understanding the input from generating an appealing answer. Clarification research treats detecting the need to ask, selecting an informative question, and responding after clarification as distinct capabilities; current language models continue to struggle with when clarification is warranted (Testoni and Fernández, 2024; Zhang and Choi, 2025). Framing clarification as an expression of calibrated uncertainty connects this to work on estimating and linguistically ex-

pressing uncertainty in language models (Huang et al., 2024; Lin, 2024): a model that cannot signal uncertainty on ambiguous input will answer rather than ask. Such a framework must (i) hold the underlying concern constant across register and language, (ii) score concern interpretation, risk/action, and clarification need separately, (iii) expose under-triage and unsafe advice rather than rely on aggregate accuracy, and (iv) prevent overlap between evaluation and adaptation families. Our controlled suite is positioned around these requirements. Native-speaker review targets semantic and register preservation in Arabic and French, while source- and paraphrase-family exclusions separate evaluation from training. The blinded clinical-rating designs used in PCOS and fertility studies motivate independent review of response quality (Graca et al., 2026; Gély et al., 2026), but our primary understanding measures remain tied to predefined item labels. Because LLM judges can show only moderate agreement and systematic rubric biases (Donati et al., 2025), rubric-scored cultural sensitivity, helpfulness, clarity, and safety are reported separately from deterministic label metrics and should be calibrated against independently reviewed responses.

3 Methods

3.1 Benchmark construction

HerHealthEval is built on existing female-health resources rather than free-form synthesis. Seed questions are drawn from menstrual-health QA (MENST/MenstLLaMA) and women’s-health cases identified in patient–doctor dialogue corpora (MedDialog/HealthCareMagic/iCliniq), filtered to three concern categories: **menstrual**, **PCOS/hormonal**, and **fertility**. This yields 90 seed conditions (30 per category).

Each seed is a fixed underlying clinical state expressed in six *communication registers*: *canonical*, *clinical*, *layperson*, *indirect/cultural*, *ambiguous*, and *emotionally concerned*. For example, the same amenorrhea seed appears as “My menstrual cycle has become irregular” (clinical), “My periods are all over the place lately” (lay), and “Something feels wrong with my cycle” (ambiguous). Holding the clinical state fixed while varying only the surface form is what lets us attribute differences in model behavior to *language understanding* rather than to differences in the underlying case. This gives $90 \times 6 = 540$ items per lan-

guage.

Every item carries gold labels for concern **category**, **risk level**, **recommended action**, and whether the input genuinely **requires clarification**. Labels are a property of the seed and are therefore identical across languages and registers, which is what makes cross-language and cross-style consistency measurable.

3.2 Translation and validation protocol

The English benchmark is translated into French and Arabic. Machine translation provides an initial draft, which is then validated in two tiers, following the supervisor’s protocol: (1) a *native speaker* reviews fluency and naturalness, and (2) a *linguistic professional* reviews meaning preservation, ambiguity, and cultural appropriateness (critical for the indirect/cultural and ambiguous registers, where literal translation would destroy the phenomenon under study). Per-item validation status is recorded in the benchmark. This produces 1,620 parallel, human-validated items (540×3 languages). Adaptation (fine-tuning) data is drawn from a disjoint seed pool and passes an explicit leakage filter (verbatim-question and shared-answer overlaps with the benchmark are removed before training).

3.3 Models

Following the comparison design, we evaluate: (i) a **base multilingual LLM** — Qwen3.5-9B, zero-shot (M2); and (ii) a **fine-tuned multilingual LLM** — QLoRA adaptation of the same model on the joint English+French+Arabic corpus. Qwen3.5-9B is natively multilingual, so it serves as both the base and multilingual-base reference. Retrieval augmentation is left to future work.

Fine-tuning uses QLoRA (4-bit NF4; LoRA rank 16, $\alpha = 16$, on all attention and MLP projections), learning rate 2×10^{-4} with cosine schedule and 0.03 warmup, max gradient norm 0.3, bf16, paged 8-bit AdamW, two epochs, effective batch size 16, max sequence length 2048, seed 3407, with loss computed on responses only and thinking-mode disabled so training and evaluation prompts match. Targets are wrapped in the exact structured schema the benchmark scores (predicted category, interpreted symptom, risk, recommended action, clarification decision, unsafe flag, response text). To keep supervision language-independent, risk and clarification labels are derived from the English source and attached to

French/Arabic rows by seed identifier, avoiding a cross-lingual labeling failure we document in Section 4.2.

3.4 Evaluation metrics

We evaluate *language understanding*, not answer generation. All metrics are computed deterministically from the model’s structured prediction against gold:

- **Symptom-interpretation accuracy:** predicted vs. gold concern category (majority baseline 0.333 on the balanced multilingual set).
- **Under-triage / unsafe-reassurance rate:** $P(\text{predict routine} \mid \text{gold see-doctor})$ — the safety-critical error.
- **Clarification appropriateness:** recall on genuinely ambiguous items and specificity on the rest.
- **Cross-language consistency:** for a fixed (seed, style), whether the risk/category verdict agrees across English, French, and Arabic (operationalizes meaning preservation).
- **Cross-style consistency:** whether the verdict is stable across the six registers of the same seed (surface-form robustness).
- **Misunderstanding rate:** complement of interpretation accuracy, with a strict variant that counts unparseable outputs.

Because gold risk is uniformly see-doctor, we report per-class recall, under-triage, and majority baselines rather than aggregate accuracy alone. Paired model comparisons use McNemar’s test on the shared items; rate estimates report bootstrap 95% confidence intervals. Cultural sensitivity is proxied by performance on the indirect/cultural register; automatic scoring of response helpfulness/clarity is left to future work (Limitations).

4 Results

We report two complementary evaluations. Section 4.1 is an English-only pilot that isolates how supervised fine-tuning reshapes structured triage behavior (target-format sensitivity and clarification collapse). Section 4.2 is the main multilingual evaluation on the aligned HerHealthEval benchmark (English, French, Arabic on the

Metric	M2	M3 _{QA}	M3 _J	M3 _{J+O}	Base
parse_ok	1.000	0.865	0.998	0.996	—
risk accuracy	0.667	0.625	0.623	0.665	0.644
under-triage ↓	0.718	0.800	0.647	0.605	—
clarif. recall	0.856	0.174	0.044	0.000	—
category accuracy	0.539	0.499	0.549	0.550	—
consistency (risk)	0.644	0.456	0.433	0.444	—
consistency (cat.)	0.156	0.189	0.344	0.378	—

Table 1: English benchmark results ($n = 540$). M2 = zero-shot baseline; M3_{QA} = QLoRA fine-tune on plain question-answer pairs; M3_J = fine-tune on eval-shaped structured (JSON) targets; M3_{J+O} = structured targets with $4\times$ oversampling of the clarification (ambiguous-style) training rows. “Base” = majority-class baseline. Arrows: lower is better for under-triage. Best per row in **bold**. Consistency = fraction of seed groups given the same label across all six styles.

same seeds), which measures cross-language consistency and cross-lingual under-triage. Gold labels are skewed toward see-doctor, so throughout we report majority-class baselines alongside accuracy and treat per-class recall, under-triage, and clarification recall as the honest headlines.

4.1 English pilot: fine-tuning reshapes triage behavior

The pilot uses a frozen English benchmark ($n = 540$: 90 seeds $\times$ 6 communication styles; risk majority baseline 0.644, clarification majority baseline 0.833).

4.1.1 Zero-shot baseline (M2)

Table 1 reports the zero-shot baseline (M2), an instruction-tuned Qwen3.5-9B prompted with our structured triage schema. M2 parses at 100% and reaches 0.667 risk accuracy, only marginally above the 0.644 majority baseline. The safety-critical weakness is *under-triage*: of 174 items whose gold risk is see-doctor, M2 routes 71.8% to routine (per-class recall for see-doctor is only 0.264, versus 0.897 for routine). M2 does, however, retain strong clarification-seeking: recall of 0.856 on the 90 items that require clarification, at 0.862 specificity. Category accuracy is 0.539, with fertility weakest (recall 0.467). The model gives a consistent risk label to only 64.4% of seed groups across their six phrasings, and a consistent category to just 15.6%, direct evidence that surface communication style, not clinical content, drives much of the variance in interpretation.

4.1.2 Fine-tuning is highly sensitive to target format

We fine-tune Qwen3.5-9B with QLoRA on leakage-controlled adaptation data drawn from the same seed pool as the benchmark but disjoint from it (verbatim-question and shared-answer overlaps are removed before training). We compare two target formats. $M3_{QA}$ trains on plain question→answer text; $M3_{JSON}$ trains on targets wrapped in the exact structured schema the benchmark scores.

Target format dominates the outcome. Training on plain answers ($M3_{QA}$) degrades every structured behavior: JSON parse compliance falls to 0.865, under-triage *worsens* to 0.800, and clarification-seeking collapses from 0.856 to 0.174, because the plain-answer targets never model the structured triage act the benchmark measures. Reformatting the identical training content into the eval schema ($M3_{JSON}$) restores parse compliance to 0.998, reduces under-triage below the zero-shot baseline to 0.647, and more than doubles cross-style category consistency (0.156 to 0.344). By McNemar’s paired test on the 540 items, $M3_{JSON}$ differs from $M2$ significantly on risk ($p=0.03$, favoring $M2$) but not on category ($p=0.71$) or clarification ($p=0.30$) correctness; parse compliance is statistically unchanged ($p=1.0$).

4.1.3 Clarification-seeking does not survive fine-tuning

Adding structured targets recovers format but not clarification: $M3_J$ asks for clarification on almost none of the 90 items that require it (recall 0.044). We tested whether this is a simple data-balance problem by oversampling the ambiguous (clarification) training rows four-fold, raising them from 10.5% to 32% of the mixture ($M3_{J+O}$). Oversampling further improves triage— $M3_{J+O}$ matches the zero-shot baseline on risk accuracy (0.665 vs. 0.667), attains the lowest under-triage of any model (0.605), and the highest cross-style category consistency (0.378)—but clarification recall does not recover; it falls to 0.000. Every fine-tuned variant, regardless of format or clarification-oversampling, drives clarification-seeking to near zero while pushing specificity toward 1.0: the adapted models simply stop asking and always answer. Clarification-seeking on genuinely ambiguous input thus behaves as a fragile, emergent instruction-following capability of the base model

that supervised fine-tuning on answer-style data robustly erases, and that naive rebalancing of the training mixture does not restore. By McNemar’s test, $M3_{J+O}$ is statistically indistinguishable from $M2$ on risk ($p=0.92$), category ($p=0.70$), and clarification ($p=0.20$) correctness, and on parse compliance ($p=0.48$); its gains are concentrated in the safety-oriented rate metrics (under-triage, over-triage) and cross-style consistency rather than in raw accuracy.

4.2 Multilingual evaluation on HerHealthEval

We now evaluate on the aligned multilingual benchmark (benchmark_multilingual_v1): the same 90 *menst-** seeds rendered in English, French, and Arabic across the six styles ($n = 540$ per language, 1,620 parallel items), with identical gold labels across languages. Because every seed appears in all three languages, this set supports *cross-language consistency* and *meaning-preservation* measurement, which the English pilot cannot. Category is balanced (180/180/180; majority baseline 0.333); risk is uniformly see-doctor (under-triage is the honest headline); 24/540 items require clarification.

We compare the base multilingual model ($M2$, Qwen3.5-9B zero-shot) against the fine-tuned multilingual model. To isolate the effect of *label quality* we report two fine-tunes: $M3_{ml_{v1}}$, whose non-English risk labels were corrupted by an English-only risk heuristic applied to translated answers (mislabeling 99.8% of French/Arabic training rows routine), and $M3_{ml_{v2}}$, which derives risk from the English source by *row_id* so labels are language-independent.

4.2.1 Cross-lingual under-triage from language-dependent labels

Our clearest “why models misunderstand non-English” result is a controlled ablation. $M3_{ml_{v1}}$ *understands* French: on the French benchmark it parses at 0.998 and reaches category accuracy 0.605 (higher than its English category accuracy), yet it routes 539/540 French items to routine (under-triage ≈ 1.0) — it almost never recommends seeing a doctor in French. The identical pattern holds for Arabic (parse 0.998, category accuracy 0.618, under-triage 0.998), and the French and Arabic predictions agree on the (collapsed) risk verdict for 99.4% of seeds ($n = 540$): the model under-triages non-English input *systemati-*

Model	Lang	interp. acc	under-triage ↓	clar. recall
M2 (base)	EN	0.617	0.463	0.208
	FR	0.614	0.440	0.333
	AR	0.617	0.420	0.375
M3ml _{v1}	FR	0.605	1.000	0.000
	AR	0.618	0.998	0.000
M3ml _{v2}	EN	0.644	0.439	0.000
	FR	0.617	0.450	0.000
	AR	0.646	0.443	0.000

Table 2: Multilingual results on HerHealthEval ($n = 540/\text{language}$). interp. acc = category (symptom-interpretation) accuracy; under-triage = fraction of gold see-doctor routed to routine (lower is better); clar. recall on the 24 gold-yes items. M3ml_{v1} vs. M3ml_{v2} isolates the effect of language-independent risk labeling: M3ml_{v1} collapses non-English under-triage to ≈ 1.0 , while M3ml_{v2} restores it to the base model’s level and attains the best interpretation accuracy in every language. (M3ml_{v1} English uses a separate legacy set and is omitted.)

cally and consistently, not sporadically. The cause is not comprehension but supervision: the fine-tuning pipeline derived risk labels by matching *English* consult words against the *translated* answers, so 99.8% of French (and Arabic) training rows were labeled routine. Recovering risk from the English source by row_id (M3ml_{v2}) restores the training risk distribution to match English (routine 0.58 / see-doctor 0.40), and this single change repairs the behavior: M3ml_{v2} under-tries 0.450 of French and 0.443 of Arabic see-doctor items — back to the base model’s 0.44/0.42 and far below M3ml_{v1}’s ≈ 1.0 — while its category interpretation is statistically unchanged from M3ml_{v1} (§4.2.3). This demonstrates that cross-lingual safety failures can arise from *language-dependent label quality* rather than from the model’s inability to understand the non-English input — a failure mode invisible to English-only evaluation, and one fixed by correcting supervision rather than by adding data or languages.

4.2.2 Cross-language consistency

Because every seed appears in all three languages with identical gold labels, we can measure whether a model gives the *same* verdict across languages on the same case. For the base model M2, category (interpretation) verdicts agree across English, French, and Arabic on 82.4% of seeds and risk verdicts on 71.1% ($n = 540$ aligned triples). This is genuine consistency over a non-degenerate risk distribution (base under-triage 0.42–0.46 across the three languages): the same case is interpreted

largely language-invariantly, and the residual $\approx 18\%$ category and $\approx 29\%$ risk disagreement is exactly the cross-lingual instability the benchmark is designed to expose.

By contrast, M3ml_{v1} shows near-perfect FR $\leftrightarrow$ AR risk “consistency” (0.994), but this is an *artifact* of both languages collapsing to routine — consistency around an unsafe constant, not around a correct decision. The comparison is instructive: M2’s *lower* consistency (0.711) is the more trustworthy number because it reflects a live risk judgment, whereas M3ml_{v1}’s higher number reflects the absence of one. M3ml_{v1} category consistency (0.880) remains close to M2’s (0.824), confirming that the fine-tune left cross-lingual condition understanding intact and degraded only the risk mapping.

The corrected fine-tune M3ml_{v2} behaves like the base model rather than M3ml_{v1}: risk consistency 0.681 and category consistency 0.837 ($n = 540$), both over a live risk distribution (under-triage 0.44–0.45 across languages). Its category consistency is in fact the highest of the three models, so the correction makes interpretation the *most* language-invariant while keeping risk consistency honest rather than collapsed.

4.2.3 Paired significance (McNemar)

We test M2 against M3ml_{v1} on the shared 540 items per language with McNemar’s paired test (Table 3). On *risk* the difference is both enormous and one-directional: M2 recovers the correct see-doctor verdict on 289 French and 298 Arabic items where M3ml_{v1} fails, while M3ml_{v1} is correct on *zero* items where M2 fails ($p < 10^{-63}$ both languages). The discordance has no offsetting direction — statistical confirmation that the fine-tune’s cross-lingual under-triage is systematic, not noise. On *category* (interpretation) the two models are statistically indistinguishable (FR $p = 0.75$, AR $p = 0.93$), confirming that adaptation neither helped nor harmed which condition the model recognizes. On *clarification*, raw correctness favors M3ml_{v1} (discordant pairs 9:72 FR, 10:77 AR in its favor), but only because M3ml_{v1} never asks for clarification and 95.6% of gold items require none: always-abstain wins the majority class. M2 trades that specificity for non-zero clarification recall (0.33–0.38 vs. M3ml_{v1}’s 0.00). The apparent “win” for the fine-tune on clarification accuracy is thus a majority-baseline artifact, and is reported here precisely to expose it.

Lang	Measure	b	c	p
FR	risk	289	0	2.2×10^{-64}
	category	78	73	0.75
	clarification	9	72	5.6×10^{-12}
AR	risk	298	0	2.5×10^{-66}
	category	73	73	0.93
	clarification	10	77	1.5×10^{-12}

Table 3: McNemar’s paired test, M2 (base) vs. M3ml_{v1}, on the shared 540 items per language. b = items M2 gets right and M3ml_{v1} wrong; c = the reverse. Risk discordance is entirely one-directional ($c = 0$): M2 is safer wherever the two disagree. Category shows no significant difference; the clarification column favors M3ml_{v1} only through the always-abstain majority baseline (gold is 95.6% “no clarification”).

Correcting the risk labels to be language-independent (M3ml_{v2}) reverses this picture (Table 4). Against M3ml_{v1}, M3ml_{v2} recovers the correct risk verdict on 292 French and 296 Arabic items while M3ml_{v1} is correct on *zero* items where M3ml_{v2} fails ($p < 10^{-64}$) — the same total, one-directional discordance as the base-model comparison, now in the corrected model’s favor, and with category interpretation left unchanged (FR $p = 0.42$, AR $p = 0.055$). Against the base model M2, M3ml_{v2}’s risk decisions are now statistically *indistinguishable* in the non-English languages (FR $p = 0.88$, AR $p = 0.94$) and modestly safer in English ($p = 0.011$): the corrected fine-tune has climbed all the way back to the base model’s safety rather than merely improving on M3ml_{v1}. Interpretation remains at least on par with the base model in every language, and M3ml_{v2} attains the best category accuracy in Table 2. The one measure that does not recover is clarification: like M3ml_{v1}, M3ml_{v2} rarely asks (recall 0.00), so M2 retains a genuine advantage there ($p \sim 10^{-11}$). Restoring calibrated clarification, left aside in this iteration, is the clearest remaining target.

5 Discussion

Understanding is necessary but not sufficient for safety. Our clearest result separates two things that answer-accuracy conflates: whether a model *understands* a female-health message, and whether it *acts safely* on it. The fine-tuned model interprets French and Arabic input well (category accuracy 0.61–0.62, parse compliance 0.998) yet routes almost every non-English case to routine. Comprehension is intact; the failure is in the mapping from understanding to a safe recommenda-

Comparison	Lang	Measure	b	c	p
M3ml _{v1} vs M3ml _{v2}	FR	risk	0	292	5.0×10^{-65}
		category	16	22	0.42
	AR	risk	0	296	6.7×10^{-66}
		category	16	30	0.055
M2 vs M3ml _{v2}	EN	risk	63	96	0.011
	FR	risk	84	87	0.88
	AR	risk	83	81	0.94

Table 4: McNemar’s paired test for the correction. Top: M3ml_{v1} vs. M3ml_{v2} (b = M3ml_{v1} right and M3ml_{v2} wrong; c = reverse). Correcting the labels flips the risk discordance to entirely one-directional in M3ml_{v2}’s favor ($b = 0$), while category is unchanged. Bottom: M2 (base) vs. M3ml_{v2}; after the fix, risk is statistically indistinguishable from the base model in French and Arabic. M3ml_{v1}’s English predictions use a separate legacy set and are excluded from the top block.

tion. Evaluations that score only the final answer, or that aggregate over languages, would miss this dissociation entirely.

Language-dependent label quality is a hidden multilingual failure mode. The cross-lingual under-triage we observe does not originate in the model but in *supervision*: an English-centric risk-labeling heuristic, applied to translated answers, mislabeled 99.8% of French and Arabic adaptation targets as low-risk, and the model faithfully learned to under-triage non-English input. Because French and Arabic collapse identically (risk agreement 0.994), the failure is systematic, not noise. This has a concrete methodological implication for multilingual safety benchmarks: seeds must be aligned across languages, safety metrics must be reported *per language*, and the *provenance of labels* must be audited for language independence. A benchmark that is English-only, or that reports a single pooled number, cannot surface this class of bug.

Clarification-seeking is a fragile capability that adaptation erases. The English pilot shows that supervised fine-tuning on answer-style data drives clarification-seeking on genuinely ambiguous input from a strong baseline (0.856 recall) to near zero, and that naive rebalancing of the training mixture does not restore it. In a high-stakes domain where much female-health language is indirect or vague, the ability to *ask rather than answer* is itself safety-relevant. Recovering it—via explicit clarification supervision, uncertainty-aware objectives, or retrieval-grounded abstention—is a natural direction for future work, alongside re-

trieval augmentation for grounded interpretation.

Limitations

This initial study covers three symptom categories and three languages and therefore does not represent the full diversity of women's-health concerns, dialects, cultures, or care settings. Native-speaker validation assesses translation quality and preservation of register; it is not a substitute for clinical validation of every gold target. The current risk and clarification labels are clinically motivated but require further expert review, and their class imbalance limits conclusions drawn from aggregate accuracy. Finally, model outputs are evaluated as evidence of linguistic understanding in a controlled research setting, not as medical advice or proof of readiness for clinical deployment.

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